

Foundations of Motivic Galois Theory: An Interim Working Paper

Toward a Tannakian Framework Unifying Cohomological Symmetries of Algebraic Varieties

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Abstract

This note lays the conceptual and structural groundwork for the problem of a *motivic Galois theory*. Beginning with classical Galois theory of number fields and the Tannakian reconstruction of affine group schemes from their representation categories, we review the multiplicity of cohomology theories attached to a smooth projective variety, the comparison isomorphisms relating them, and the periods that quantify these comparisons. We then sketch Grothendieck's vision of a universal cohomology theory through which all realizations factor, the resulting category of motives, and the pro-algebraic group G_{mot} obtained as the Tannakian automorphism group of the Betti fiber functor. We close by formulating four open problems that, taken together, constitute the present scope of the subject: construction of the abelian category of mixed motives, the period conjecture, computation of motivic Galois groups in concrete cases, and the position of the absolute Galois group inside its motivic counterpart. The note is preparatory; technical developments are deferred to subsequent drafts.

1. Classical Galois Theory as a Starting Point

For a Galois extension L/K , the group $\text{Gal}(L/K)$ encodes the symmetries among the algebraic relations satisfied by elements of L over K . The central object in modern arithmetic is the absolute Galois group of the rationals,

$$G_{\mathbf{Q}} := \text{Gal}(\overline{\mathbf{Q}}/\mathbf{Q}),$$

a profinite group whose continuous representations control vast portions of number theory. A first organising principle, fundamental to what follows, is that *a group can be recovered from its category of representations*. This is the content of Tannaka–Krein duality, and it is the technical engine that allows us to define Galois-type groups even when no group is initially in sight.

2. The Tannakian Formalism

A *neutral Tannakian category* over a field k is, in essence, a k -linear abelian rigid tensor category $\mathbf{T}$ equipped with an exact, faithful, tensor-preserving *fibre functor* $\omega: \mathbf{T} \rightarrow \text{Vec}_k$. The fundamental theorem of Saavedra–Deligne asserts a tensor equivalence

$$\mathbf{T} \simeq \text{Rep}_k(G), \quad G := \underline{\text{Aut}}^{\otimes}(\omega),$$

where G is an affine group scheme over k , reconstructed as the tensor-automorphisms of the fibre functor. The slogan: *specify a category with the right formal structure, and a group emerges*. This is precisely the strategy adopted in motivic Galois theory: the category will be that of motives, the fibre functor a chosen cohomological realization.

3. The Cohomology of Algebraic Varieties

Let X be a smooth projective variety over $\mathbf{Q}$. Several cohomology theories coexist:

- **Betti cohomology:** $H_{\mathbf{B}}^*(X(\mathbf{C}), \mathbf{Q})$, the singular cohomology of the complex points.
- **Algebraic de Rham cohomology:** $H_{\text{dR}}^*(X/\mathbf{Q})$, the hypercohomology of the algebraic de Rham complex $\Omega_{X/\mathbf{Q}}^{\bullet}$.

- **■-adic étale cohomology:** $H_{\text{ét}}^*(X_{\mathbf{Q}^\times}, \mathbf{Q}_\square)$, carrying a continuous action of $G_{\mathbf{Q}}$.

These theories are linked by canonical *comparison isomorphisms*; the Betti–de Rham comparison reads

$$H_B^*(X(C), \mathbf{Q}) \otimes_{\mathbf{Q}} \mathbf{C} \xrightarrow{\sim} H_{dR}^*(X/\mathbf{Q}) \otimes_{\mathbf{Q}} \mathbf{C}.$$

Each realization yields the same Betti numbers, the same Hodge numbers (in the appropriate sense), and the same L -functions. The coherence among these distinct constructions is the principal phenomenon that motivic theory seeks to explain.

4. Periods

Choosing $\mathbf{Q}$ -rational bases on each side of the Betti–de Rham comparison produces a matrix of complex numbers, the *periods* of X . Representative examples:

- $X = \mathbf{G}_m$: the period $\int_{|z|=1} dz/z = 2\pi i$.
- X an elliptic curve: the periods ω_1, ω_2 of an invariant differential.
- $X = \mathbf{P}^1 \setminus \{0, 1, \infty\}$: iterated integrals giving $\zeta(n)$ and, more generally, multiple zeta values.

These numbers occupy a delicate position between the algebraic and the transcendental. The empirical and conjectural relations among them strongly suggest the action of a hidden symmetry group — the motivic Galois group, to be defined below.

5. Grothendieck's Vision: Motives

Grothendieck proposed the existence of a *universal* cohomology theory through which all Weil cohomologies factor. Schematically, one seeks:

- a category $\mathbf{M}(k)$ of motives over k ;
- a contravariant functor $h: \text{Var}(k) \rightarrow \mathbf{M}(k)$;
- for each cohomology theory H_{γ}^* , a *realization functor* $\mathbf{M}(k) \rightarrow (\text{target category})$ through which H_{γ}^* factors.

For **pure motives** (X smooth projective), one builds objects from pairs (X, p) with p an idempotent algebraic correspondence; morphisms are correspondences modulo a chosen equivalence relation (rational, homological, numerical). The choice is consequential: only under the *standard conjectures* does one obtain an abelian, semisimple, neutral Tannakian category. For **mixed motives** (general X) the desired abelian category $\text{MM}(k)$ remains conjectural, although Voevodsky's triangulated category $\text{DM}(k)$ provides a robust working substitute.

6. The Motivic Galois Group

Granting $\mathbf{M}(k)$ as a neutral Tannakian category and taking the Betti realization ω_B as fibre functor, the Tannakian formalism produces

$$G_{\text{mot}}(k) := \underline{\text{Aut}}^{\otimes}(\omega_B),$$

a pro-algebraic group over $\mathbf{Q}$ — the *motivic Galois group*. Its expected features include:

- every motive M yielding a representation $G_{\text{mot}} \rightarrow \text{GL}(\omega_B(M))$;
- the classical Galois group $G_{\mathbf{Q}}$ appearing as a quotient or component (e.g. as π_0 of G_{mot});
- for the subcategory of mixed Tate motives over $\mathbf{Z}$, an explicit description intimately linked to multiple zeta values (Deligne, Goncharov, Brown).

7. The Problems, Sharply Stated

We may now state precisely what the phrase *the problem of motivic Galois theory* denotes in the present working note.

(a) Constructing the category.

Realize $\mathbf{M}(k)$ — pure or mixed — as an abelian, neutral Tannakian category. For pure motives this requires the standard conjectures (Künneth, Hodge-type, ...). For mixed motives, even a correct t -structure on $\mathrm{DM}(k)$ is open.

(b) The period conjecture.

Grothendieck's period conjecture asserts that for a motive M over $\mathbf{Q}$, the transcendence degree over $\mathbf{Q}$ of the field generated by the periods of M equals the dimension of the motivic Galois group of M . This is a sweeping generalization of Lindemann–Weierstrass and remains open even in apparently elementary cases — for instance, the algebraic independence of π and $\zeta(3)$.

(c) Computing G_{mot} .

Even granting the framework, identifying G_{mot} for specific motives or Tannakian subcategories is deep. Successes (Tate motives, certain hypergeometric motives) have driven activity at the interface of arithmetic, mathematical physics (Feynman integrals), and the combinatorics of Hopf algebras of multiple zeta values.

(d) Galois action and arithmetic.

How does $G_{\mathbf{Q}}$ sit inside G_{mot} ? The Mumford–Tate, Tate, and Hodge conjectures may all be read as predictions about which $\mathbf{Z}$ -adic or Hodge-theoretic structures genuinely originate from algebraic geometry — that is, which structures are *motivic*.

8. Synthesis

Unifying picture. Classical Galois theory is the Galois theory of zero-dimensional varieties (number fields). Motivic Galois theory is what one obtains by allowing varieties of arbitrary dimension and seeking the symmetries among *all* their cohomological invariants simultaneously. The problem, in the present state of knowledge, is that this picture is constructed only in part; the parts that work nevertheless illuminate some of the deepest questions in mathematics.

9. Next Steps for Subsequent Drafts

- Develop in detail the Tannakian reconstruction theorem and its hypotheses.
- Treat the standard conjectures and their role in the construction of pure motives.
- Survey Voevodsky's $\mathrm{DM}(k)$ and the conjectural t -structure.
- Present in full Brown's theorem on mixed Tate motives over $\mathbf{Z}$ and the implications for multiple zeta values.
- Formulate working conjectures and computational targets for periods of selected hypergeometric motives.

Indicative References

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End of interim working paper. Comments and corrections are welcomed prior to circulation of v0.2.